



Alimentary system module

Liver, pancreas and biliary system

SGL 1

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Hepatitis

- A general term that describes the presence of liver parenchymal inflammation.
- Causes:
 - Infective agents, which could be:
 - Viral
 - Hepatotropic viruses like HAV, HBV, HCV, HDV, HEV.
 - Other viruses like CMV, EBV.
 - Bacterial.
 - Parasites and helminthes.
 - Drugs.
 - Toxins.
 - Autoimmune.
 - Metabolic.

Viral hepatitis

Clinicopathologic syndromes of viral hepatitis

Viral hepatitis may manifest itself in different clinical and pathologic form, these include:

1. Acute asymptomatic infection with recovery.
2. Acute symptomatic infection with recovery;
 - The signs and symptoms last <6 months.
 - It can be caused by any of the hepatitis viruses.
 - The disease follows a course consists of:
 1. An incubation period.
 2. Symptomatic pre-icteric phase: include malaise, weakness, right upper quadrant pain, nausea, anorexia.
 3. Symptomatic icteric phase: jaundice, or dark urine.
 4. Convalescence phase.

Clinicopathologic syndromes of viral hepatitis

- 3. Acute liver failure.**
- 4. Chronic hepatitis is defined as symptomatic, biochemical, or serologic evidence of continuing or relapsing hepatic disease for more than 6 months.**
- 5. Carrier state: is defined as an individual who harbors and can transmit an organism but has no symptoms, it include**
 - 1. Individuals who harbor the virus but have no liver disease.**
 - 2. individuals who harbor the virus and have asymptomatic non-progressive liver damage.**

Table shows summary of differences between the different types of viral hepatitis

Common Virus Name	Hepatitis A (HAV)	Hepatitis B (HBV)	Hepatitis C (HCV)	Hepatitis D (HDV)	Hepatitis E (HEV)
Common disease name	"Infectious"	"Serum"	"Post-transfusion" or "non-A, non-B"	"Delta"	"Enteric"
Virus	<i>Hepatovirus</i> nonenveloped capsid RNA	<i>Hepadnavirus</i> enveloped DNA	<i>Flavivirus</i> enveloped RNA	Defective enveloped circular RNA	<i>Hepevirus</i> nonenveloped capsid RNA
Transmission	Fecal-oral	Parenteral, sexual, perinatal	Parenteral, sexual	Parenteral, sexual	Fecal-oral
Severity	Mild	Occasionally severe	Usually subclinical	Co-infection with HBV occasionally severe; super-infection with HBV often severe	Normal patients: mild; pregnant patients: severe
Chronicity or carrier state	No	Yes	Yes (high)	Yes	No
Clinical diseases	Acute hepatitis	- Acute hepatitis - Chronic hepatitis - Cirrhosis - Hepatocellular carcinoma (HCC)	- Chronic hepatitis - Cirrhosis - HCC	- Acute hepatitis - Chronic hepatitis - Cirrhosis - HCC	Acute hepatitis
Laboratory diagnosis	Symptoms and anti-HAV IgM	Symptoms and serum levels of HBsAg, HBeAg, and anti-HBc IgM	Symptoms and EIA for anti-HCV	Anti-HDV ELISA	Tests not routinely available
Prevention	Vaccine, hygiene	Vaccine			Hygiene
Treatment	Supportive	Antivirals, interferons, transplant	Antivirals, interferons, transplant	See hepatitis B	Supportive

Hepatitis B Virus

- **Its an important pathogen in to liver pathology since it can produces different outcomes which include:**
 - **Asymptomatic “healthy” carrier state.**
 - **Acute hepatitis with recovery and clearance of the virus.**
 - **Non-progressive chronic hepatitis**
 - **Progressive chronic disease ending in cirrhosis.**
 - **Fulminant hepatitis with massive liver necrosis.**
 - **A precursor for the development of hepatocellular carcinoma.**

Serologic diagnosis and monitoring of Viral hepatitis B

Table 19-3. Hepatitis B Terminology and Markers

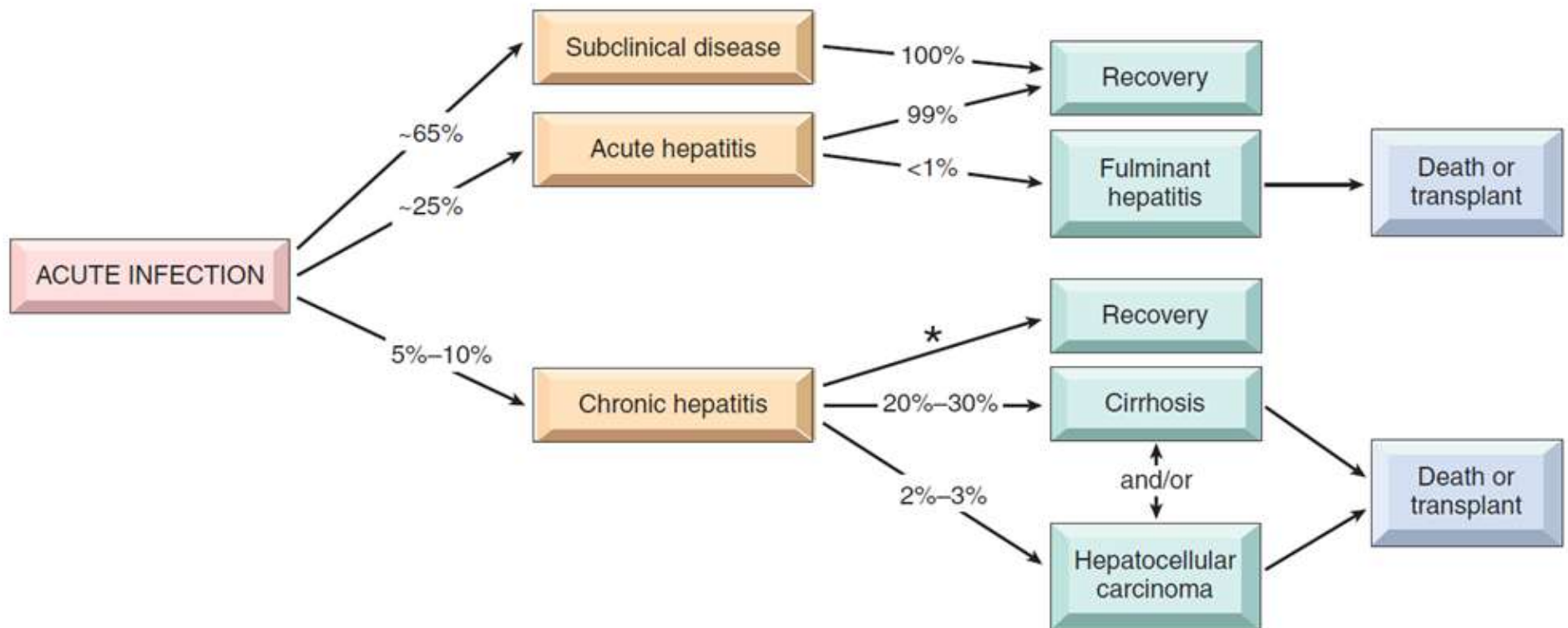
Abbreviation	Name and Description
HBV	Hepatitis B virus, a hepadnavirus (enveloped, partially doublestranded DNA virus); Dane particle = infectious HBV
HBsAg	Antigen found on surface of HBV; also found on spheres and filaments in patient's blood: positive during acute disease; continued presence indicates carrier state
HBsAb	Antibody to HBsAg; provides immunity to hepatitis B
HBcAg	Antigen associated with core of HBV
HBcAb	Antibody to HBcAg; positive during window phase; IgM HBcAb is an indicator of recent disease
HBeAg	A second, different antigenic determinant on the HBV core; important indicator of transmissibility
HBeAb	Antibody to e antigen; indicates low transmissibility
Delta agent	Small RNA virus with HBsAg envelope; defective virus that replicates only in HBV-infected cells

Interpretation of hepatitis B serology tests

HBSAG	ANTI-HBc	ANTI-HBs	INTERPRETATION
–	–	–	Susceptible (never infected)
–	IgG	IgG	Immune because of natural infection
–	–	IgG	Immune because of hepatitis B immunization
+	IgM	–	Acutely infected
+	IgG	–	Chronically infected

Anti-HBc, hepatitis B core antibody; anti-HBs, hepatitis B surface antibody; HBsAg, hepatitis B surface antigen; IgG, immunoglobulin G; IgM, immunoglobulin M; +, positive; –, negative.

Potential outcomes of hepatitis B infection in adults



Hepatitis C Virus

- The acute illness is usually less severe than HBV.
- HCV is a major cause of persistent infection and chronic hepatitis which develops in 80%-90% of patient while cirrhosis occurs in about 20%.
- Blood transfusion was the major source of transmission, however, this is markedly reduced nowadays because of the better use of screening.
- There is an increase in the transmission caused by IV opioid drugs abusers.
- One-third of individuals have no identifiable transmission risk factors.

Viral hepatitis/ Pathologic changes

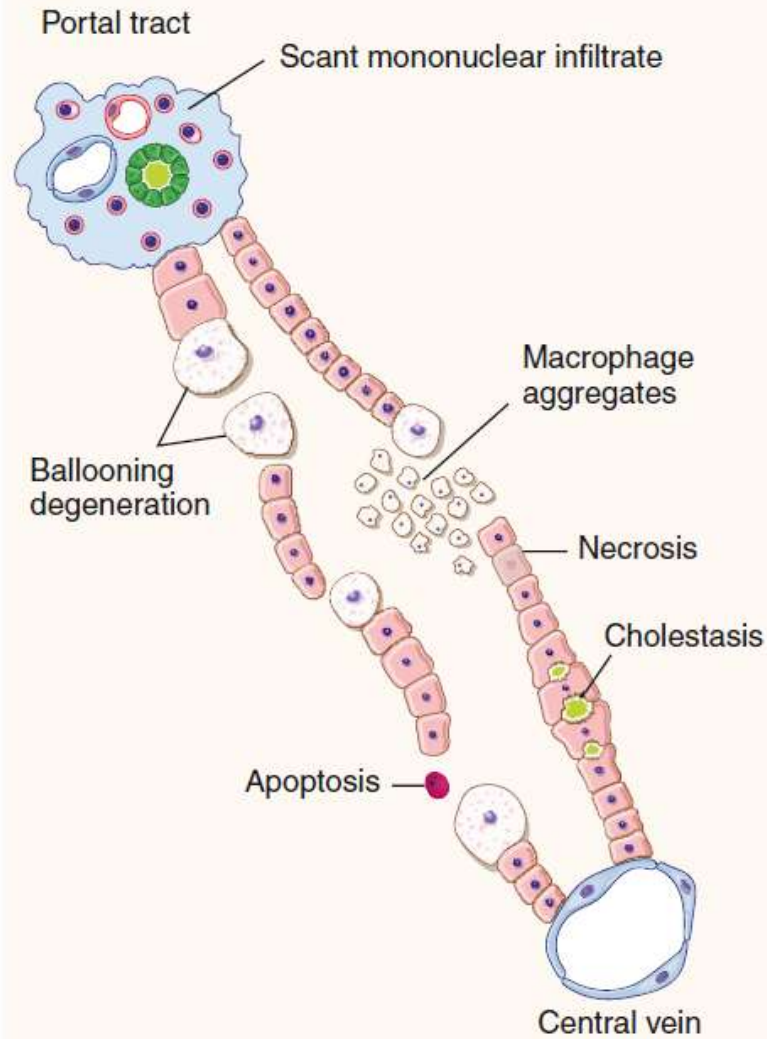
- **In acute viral hepatitis:**
- **Grossly: the liver may be normal in size or enlarged (due to inflammation), or shrunken (due to massive liver necrosis/ acute liver failure.**
- **Microscopically: the liver shows lobular disarray, hepatocyte swelling, apoptotic hepatocytes, lymphocytes in portal tracts and in the lobule, hepatocyte regeneration, and cholestasis.**

Viral hepatitis/ Pathologic changes

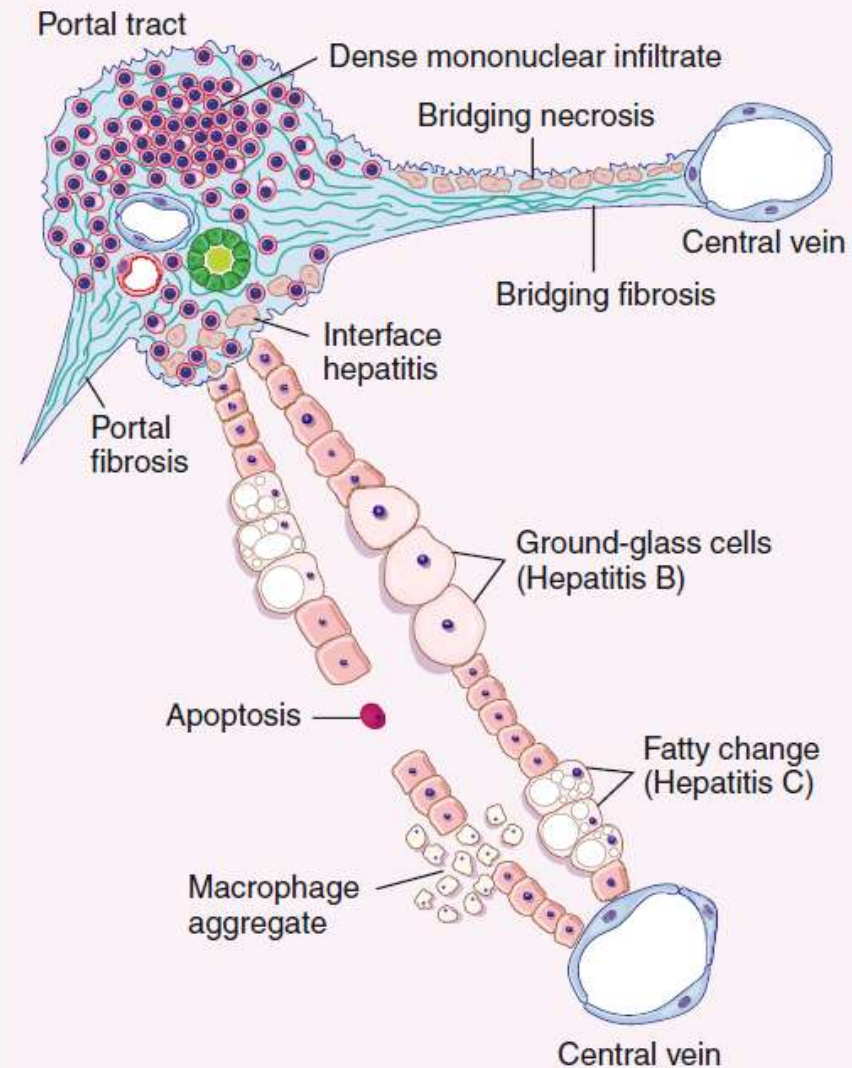
- In chronic viral hepatitis the defining histologic feature is portal lymphocytic or lymphoplasmacytic inflammation associated with fibrosis.
 - In chronic persistent hepatitis in which inflammation confined to **portal tracts**.
 - In chronic active hepatitis in which inflammation spilling into the **parenchyma**.

Viral hepatitis/ Pathologic changes

ACUTE HEPATITIS



CHRONIC HEPATITIS



Other infections:

- **Other infectious organisms can reach the liver through several pathways:**
 - **Ascending infection, via the gut and biliary tract.**
 - **Vascular seeding, most often through the portal system via the GIT.**
 - **Direct invasion, from an adjacent source (e.g., bacterial cholecystitis)**
 - **Penetrating injury (Accidental or iatrogenic).**

Other infections:

Examples include:

- **Bacteria:** like *Staph. aureus*, *Sal. typhi* and *Trep. pallidum*.
- **Entamoeba histolytica:** causes amebic liver abscess.
- **Malarial parasites**
- **Schistosomiasis.**
- **Echinococcal infections:** causes hydatid cyst of the liver.
- **Liver flukes:** a predisposing factor for cholangiocarcinoma in southeast Asia.

Autoimmune hepatitis

- **Is a chronic, progressive disorder, characterized by:**
 - **Genetic predisposition**
 - **Association with other autoimmune diseases**
 - **Presence of autoantibodies**
 - **Therapeutic responsiveness to immunosuppression.**
- **The presentation varies from asymptomatic disease, detected by elevated transaminases, to acute and chronic hepatitis.**

Autoimmune hepatitis

- The disease is of two types:
 - Type 1: is characterized by the presence of antinuclear antibodies (ANAs); anti-smooth muscle antibodies (ASMAs), and anti-soluble liver antigen/liver-pancreas antigen (anti-SLA/LP) antibodies.
 - Type 2: is characterized by anti-liver kidney microsome-1 antibodies (anti-LKM-1); anti-liver cytosol-1 (anti-LC1) antibodies; and anti-SLA/LP.

Drug- and toxin-induced liver injury

- **Liver is the major drug metabolizing and detoxifying organ in the body, hence, is subjected to injury by enormous number of therapeutic and environmental chemicals.**
- **Liver injury may be immediate or take weeks to months to develop.**
- **Liver damage may occurs by two mechanisms:**
 - **Predictable reactions: it is a dose-dependent reaction, like acetaminophen toxicity.**
 - **Unpredictable (idiosyncratic) reactions: depend on host-specific factors, e.g. reaction to chlorpromazine.**

Drug- and toxin-induced liver injury

Pattern of injury	Causative agents	
Fatty liver disease	Ethanol, steroid, aspirin	
Cirrhosis	Ethanol, methotrexate, vitamin A	
Neoplasm	Hepatocellular adenoma	Oral contraceptives, anabolic steroids
	Hepatocellular carcinoma	Alcohol, Thorotrast
	Cholangiocarcinoma	Thorotstat
	Angiosarcoma	Thorotrast, vinyl chloride
Cholestasis	Contraceptive and anabolic steroids, antibiotics, antiretroviral therapy	
Cholestatic hepatitis	Antibiotics, phenothiazines, statins	
Hepatocellular necrosis	Acetaminophen.	

Alcohol-Related Liver Disease

- “Alcohol-related liver disease” includes:
 - **Hepatic steatosis:** occur in 90-100% of chronic excessive consumers.
It is caused by the effect of ethanol which, both, increase the synthesis and impair the excretion of lipids that lead to their intracellular accumulation
 - **Steatohepatitis:** occur in 10-35% of individual, sometimes after weeks or months of excessive consumption.
Further accumulation of fat and toxic effect of ethanol induce inflammatory reaction characterized by neutrophilic infiltration.
 - **Fibrosis and cirrhosis:** occur in 8-20% of individual, usually after 15 to 20 years of chronic excessive consumption.
The ongoing cell destruction and inflammation with further progress to fibrosis and subsequent cirrhosis.

Alcohol-Related Liver Disease

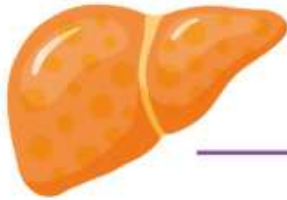
- Is a major cause of chronic liver diseases and deaths resulting from cirrhosis in Western countries.
- Hepatocellular carcinoma arises in 10% to 20% of patients with cirrhosis secondary to excess alcohol use.
- **Regression:** Even when cirrhosis is established, complete abstinence may lead, at least, to **partial regression** of scarring, parenchymal regeneration from a micronodular to macronodular cirrhotic organ and ; rarely, there is **complete regression** of cirrhosis.

The 3 stages of alcohol-related liver disease

Alcohol-related fatty liver

Often completely reversible

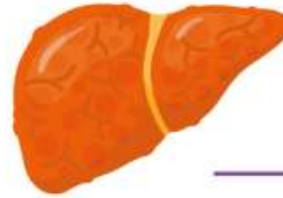
A build-up of fat in the liver.



Alcohol-related hepatitis

Sometimes reversible

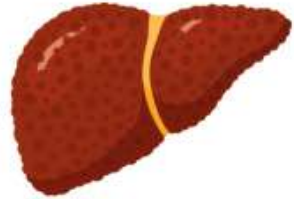
Scar tissue forms (fibrosis) and ongoing alcohol use causes inflammation in the fatty liver.



Cirrhosis

Usually irreversible, but you can often stop it getting worse

The build-up of scar tissue makes the liver hard and stops it working properly.



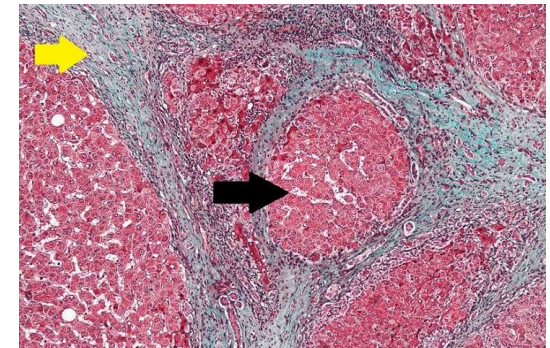
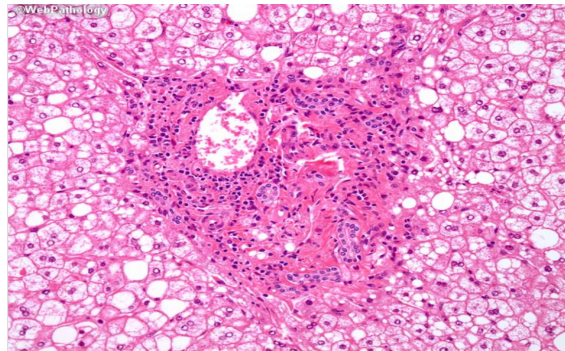
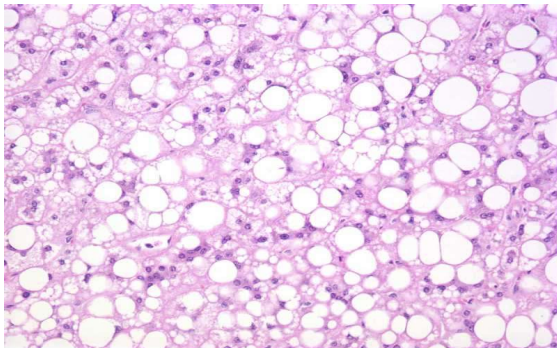
The amount of scarring (fibrosis) in your liver is linked to how serious your liver disease is.

As fibrosis gets worse it forms knots and tangles (cross-links) in the scar fibres which cannot be easily undone.

The more advanced your liver disease gets, the quicker it can worsen.

Cirrhosis can lead to liver cancer or liver failure

By finding any damage early, you could prevent it getting worse or even reverse it.



Nonalcoholic fatty liver disease (NAFLD)

- Is a common condition in which fatty liver disease develops in association with insulin resistance and the **metabolic syndrome**.
- NAFLD characterized by:
 - Type 2 DM (or family history of the condition)
 - Obesity, primarily central obesity.
 - Dyslipidemia (hyperTG, low HDL-cholesterol, high LDL-cholesterol).
 - Hypertension
- The liver can show any of the three types of changes that occur in alcohol-related liver disease (steatosis, steatohepatitis, cirrhosis).

Nonalcoholic fatty liver disease (NAFLD)

- **Nonalcoholic steatohepatitis (NASH) is the term used to describe overt clinical features of liver injury, such as elevated transaminases, and the histologic features of hepatitis.**
- **The frequency of progression from steatosis to active steatohepatitis and then to cirrhosis is low.**
- **One of the worrisome complications of NASH is the development of hepatocellular carcinoma**

Inherited metabolic liver diseases

Hemochromatosis

- Is a disease of increased levels of iron which deposits in tissue leading to tissue injury.
- It is of two types:
 - Hereditary (primary) hemochromatosis: the most common form of which is an autosomal recessive disorder of the *HFE* gene on chromosome 6p. which increases small intestine absorption of iron.
 - Acquired (secondary) hemochromatosis follow blood transfusions for chronic anemias.
- Hemochromatosis affects 5 times as many males as females.

Hemochromatosis

- **Mechanisms of liver injury include the following:**
 - **Lipid peroxidation via iron-catalyzed free radical reactions.**
 - **Stimulation of collagen formation/ fibrosis by activation of hepatic stellate cells.**
 - **DNA damage by reactive oxygen species, leading to lethal cell injury or predisposition to hepatocellular cancer.**

Hemochromatosis

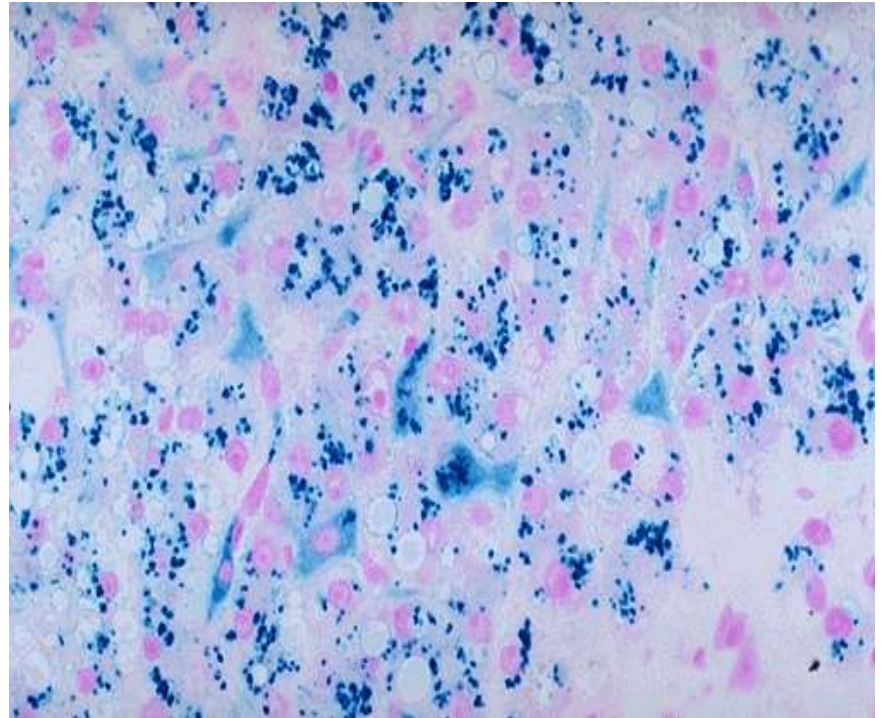
- **Excess iron can deposit in different tissue, like:**
 - **Liver:** causes micronodular cirrhosis and hepatocellular carcinoma.
 - **Skin:** causes hyperpigmentation “bronzing”
 - **Endocrine glands** caused secondary DM.
 - **Heart:** Causes heart failure and arrhythmias.
 - **Gonads:** Causes hypogonadism.
- **Diagnosis is established by**
 - **Demonstrating markedly elevated serum iron and ferritin.**
 - **Increased tissue iron levels by special stain (Prussian blue) on liver biopsy.**

Hemochromatosis

Skin pigmentation

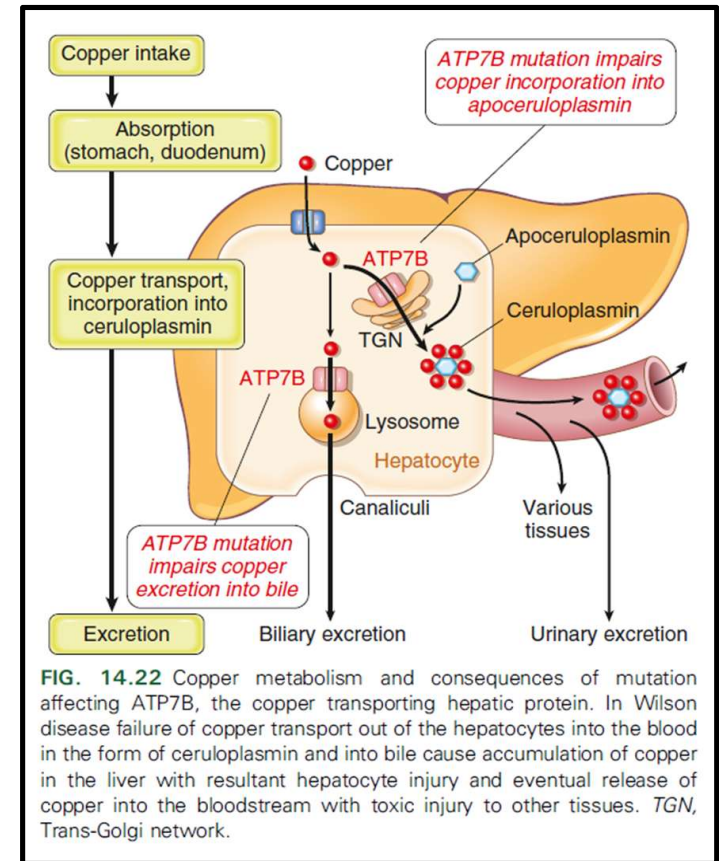


Liver biopsy stained with Prussian blue iron stain demonstrating blue granules of hemosiderin



Wilson Disease

- Is an autosomal recessive disorder caused by loss-of function mutations of the **ATP7B** gene, on chromosome 13, which results in impaired copper excretion into bile and a failure to incorporate copper into ceruloplasmin.
- Excess copper is deposit in tissues, such as the brain, corneas, kidneys, bones, joints, and parathyroid glands, causing damage.



Wilson Disease

Tissue damage occurs mostly in:

- **The liver: changes are variable and ranges from**
 - **Mild to moderate fatty change (steatosis)**
 - **Acute, fulminant hepatitis**
 - **Chronic hepatitis in Wilson disease**
 - **Steatohepatitis**
 - **Cirrhosis.**
- **The brain: primarily affects the basal ganglia.**
- **The eyes: with formation of Kayser- Fleischer rings, green to brown deposits of copper in the Descemet membrane in the limbus of the cornea.**

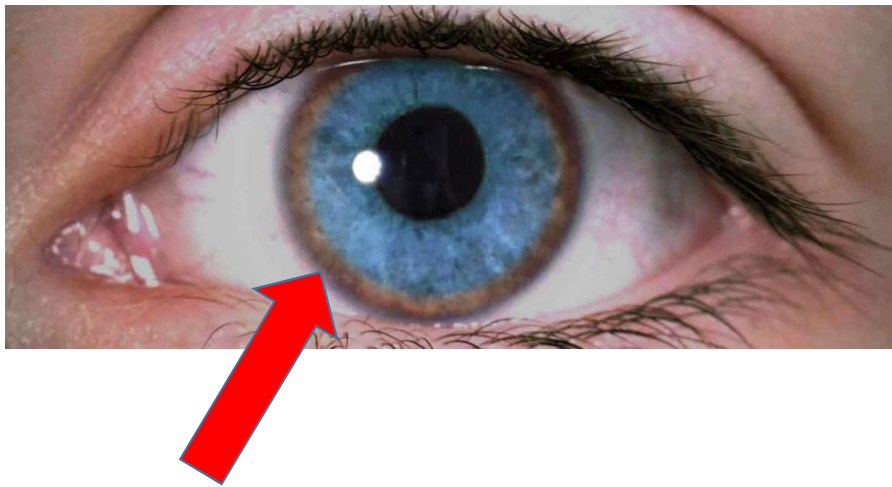
Wilson Disease

The diagnosis of Wilson disease is based on finding of:

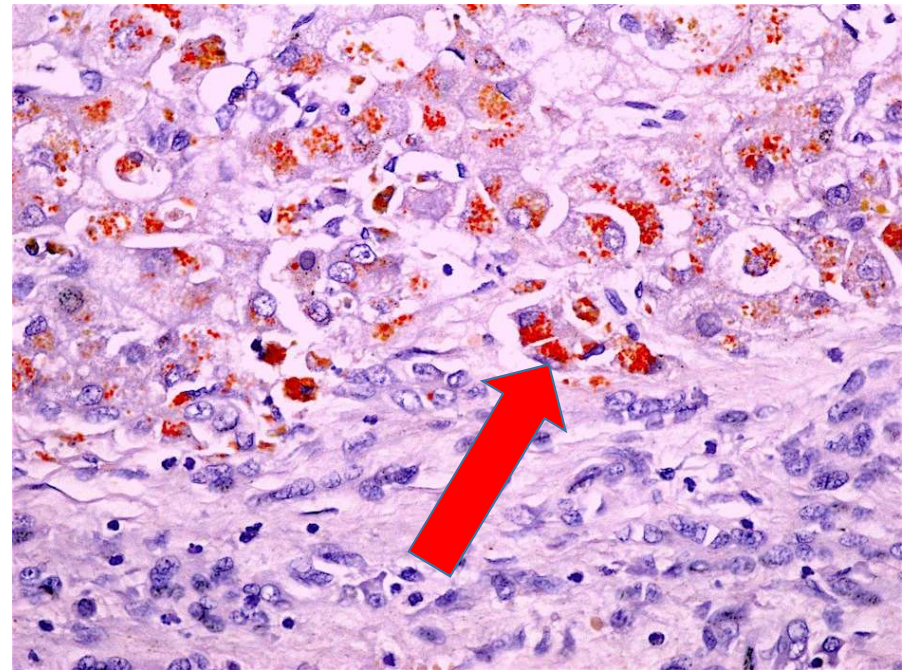
- Low levels of serum ceruloplasmin.
- Increase in hepatic copper content: an excess of 250 mg per gram dry weight of liver is considered diagnostic.
- Increased urinary excretion of copper.
- Serum copper levels is of no diagnostic value, as they may be low, normal, or elevated, depending on the stage of the liver disease.
- Genetic diagnosis is difficult because of the presence of different causative mutations in ATP7B7 gene.

Wilson Disease

Kayser-Fleischer ring



Copper stain (rhodamine) showed deposition in hepatocytes and Kupffer cells



α 1-Antitrypsin Deficiency

- **Is an autosomal recessive disorder caused by mutations, on chromosome 14, that lead to misfolding of α 1AT and loss of α 1AT function.**
- **α 1AT inhibits proteases, which are released from neutrophils at sites of inflammation, α 1AT deficiency leads to unopposed activation of proteases and subsequently destroy tissue.**

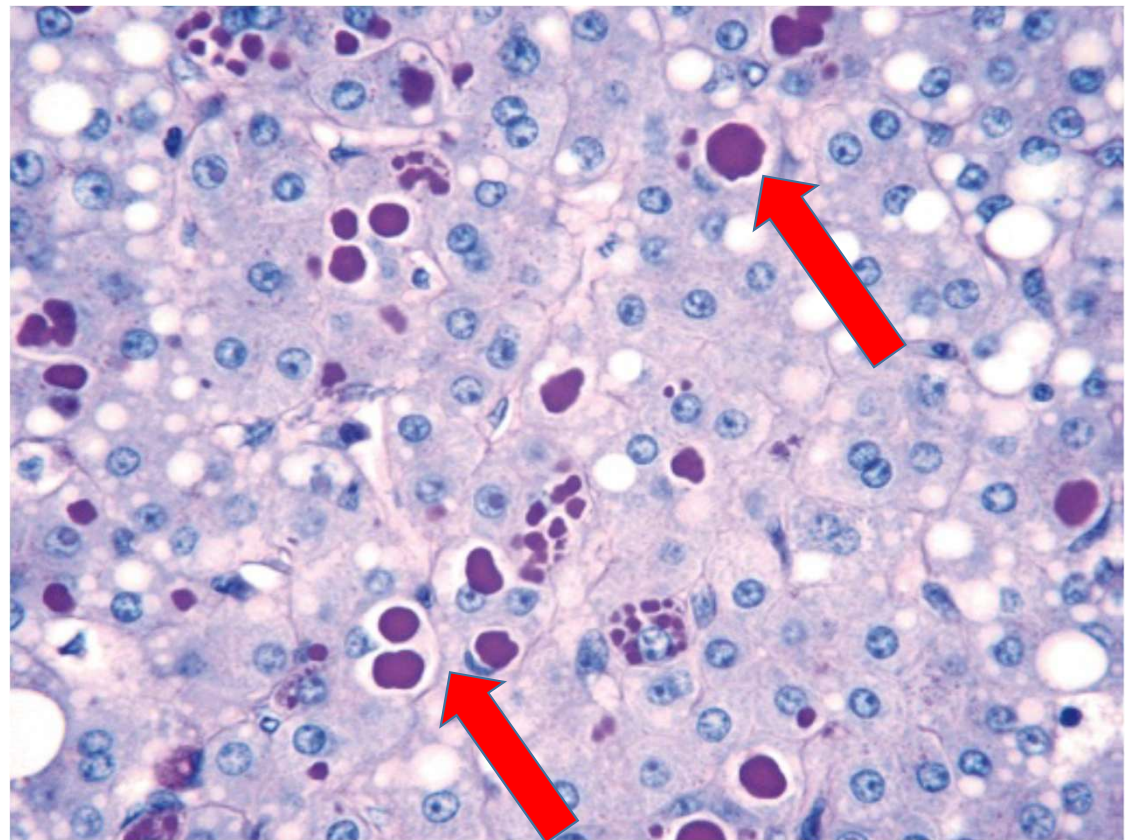
a1-Antitrypsin Deficiency

Tissue damage:

- **In liver:** is due to accumulation of abnormal a1AT protein within hepatocytes, leading to endoplasmic reticulum stress, which ultimately leads to apoptosis of hepatocytes.
Liver damage ranges from steatosis to hepatitis to cirrhosis.
- **In lungs:** a1AT deficiency leads to unopposed activation of neutrophil proteases (e.g., elastase), which destroy elastic fibers in alveolar walls, resulting in pulmonary emphysema.

α 1-Antitrypsin Deficiency

- Liver biopsy showed deposition of periodic acid Schiff (PAS) positive diastase resistant globules in hepatocytes.



References

- **Robbins & Cotran Pathologic Basis of Disease - 10th edition 2021.**
- **Robbins Basic Pathology - 11th edition 2022.**
- **Web.**